C

EST II — MATH LEVEL 2 • UNIT 2.01

Sequences, Series & Sigma Notation

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 A	Q9 D	Q10 D
Q11 A	Q12 B	Q13 D	Q14 A	Q15 A
Q16 C	Q17 A	Q18 B	Q19 D	Q20 D
Q21 B	Q22 A	Q23 D	Q24 B	Q25 B

Sequences, Series & Sigma Notation Practice Exercises

1 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

- **A** 14 ✓
- **B** 17
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

2 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

- **A** 96 ✓
- **B** 192
- **C** 98
- **D** -96

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(2)^4 = 96$. Therefore, the verified correct option is 96.

3 MCQ

Consider a geometric sequence with positive common ratio $r > 0$ where the first term is $a_1 = \frac{1}{2}$. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of r.

- **A** 2 ✓
- **B** -2
- **C** 1/2
- **D** 3

WORKED SOLUTION & VERIFICATION

The sum of the first three terms is $a_1(1 + r + r^2) = \frac{7}{2}$. Substituting $a_1 = \frac{1}{2}$ gives:

$$1 + r + r^2 = 7 \implies r^2 + r - 6 = (r + 3)(r - 2) = 0.$$

Since $r > 0$, the common ratio is $r = 2$.

4 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find a_5.

- **A** 162 ✓
- **B** 486
- **C** 165
- **D** -162

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 2(3)^4 = 162$. Therefore, the verified correct option is 162.

5 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 3$. Find a_5.

- **A** 486 ✓
- **B** 1458
- **C** 489
- **D** -486

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(3)^4 = 486$. Therefore, the verified correct option is 486.

6 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = -2$. Find a_4.

- **A** -16 ✓
- **B** 32
- **C** -18
- **D** 16

WORKED SOLUTION & VERIFICATION

$a_4 = a_1 r^{n-1} = 2(-2)^3 = -16$.

7 MCQ

A geometric sequence has $a_1 = 5$ and common ratio $r = 3$. Find a_3.

- **A** 45 ✓
- B 135
- C 48
- D -45

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 5(3)^2 = 45$. Therefore, the verified correct option is 45.

8 MCQ

A geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_5.

- **A** 81 ✓
- B -243
- C 78
- D -81

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 1(-3)^4 = 81$. Therefore, the verified correct option is 81.

9 MCQ

In an applied problem, a geometric sequence has $a_1 = 6$ and common ratio $r = 3$. Find a_5.

- A 1458
- B 489
- C -486
- **D** 486 ✓

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(3)^4 = 486$. Therefore, the verified correct option is 486.

10 MCQ

Suppose a geometric sequence has $a_1 = 5$ and common ratio $r = 3$. Find a_3.

- A 135
- B 48
- C -45
- **D** 45 ✓

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 5(3)^2 = 45$. Therefore, the verified correct option is 45.

11 MCQ

In an applied problem, a geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_6.

- **A** -243 ✓
- B -246
- C 243
- D 729

WORKED SOLUTION & VERIFICATION

$a_6 = a_1 r^{n-1} = 1(-3)^5 = -243$.

12 MCQ

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

A 12	**B 17** ✓
C 21	D 35

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is $a_n = a_1 + (n-1)d$. \n Given $a_1 = 7$ and $d = 5$, we want to find a_3. \n Substitute $n = 3$ into the formula: \n $a_3 = 7 + (3-1) \times 5$ \n $a_3 = 7 + (2) \times 5$ \n $a_3 = 7 + 10$ \n $a_3 = 17$. \n Therefore, the 3rd term is 17.

13 MCQ

Suppose a geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_4.

A 27	B 81
C -30	**D -27** ✓

WORKED SOLUTION & VERIFICATION

$a_4 = a_1 r^{n-1} = 1(-3)^3 = -27.$

14 MCQ

Find the common difference of the arithmetic sequence $6,\ 4,\ 2,\ 0, \ldots$

A -2 ✓	B 4
C 2	D -1

WORKED SOLUTION & VERIFICATION

Subtract consecutive terms: $4 - 6 = -2.$

15 MCQ

Find the common difference of the arithmetic sequence $-5,\ 3,\ 11,\ 19, \ldots$

A 8 ✓	B -8
C 9	D 3

WORKED SOLUTION & VERIFICATION

Subtract consecutive terms: $3 - (-5) = 8.$

16 MCQ

Find the common difference of the arithmetic sequence $-2,\ -8,\ -14,\ -20, \ldots$

A 6

B -5

C **-6** ✓

D -8

WORKED SOLUTION & VERIFICATION

Subtract consecutive terms: $-8-(-2)=-6$.

17 MCQ

Calculate the common difference of the arithmetic sequence $6,\ 11,\ 16,\ 21, \ldots$

A **5** ✓

B -5

C 6

D 11

WORKED SOLUTION & VERIFICATION

Subtract consecutive terms: $11-6=5$.

18 MCQ

A sequence is defined by the recurrence relation $a_{n+1}=2a_n-3$ for $n \geq 1$, with $a_1=5$. Consider a second sequence, b_n, where $b_n=a_n-k$ for some constant k. If the sequence b_n is a geometric sequence, find the value of k and the common ratio of the sequence b_n.

A k = 2, r = 3

B **k = 3, r = 2** ✓

C k = 3, r = 3

D k = -3, r = 2

WORKED SOLUTION & VERIFICATION

The first sequence is defined by $a_{n+1}=2a_n-3$, with $a_1=5$. We are given a second sequence $b_n=a_n-k$. For b_n to be a geometric sequence, the ratio $\frac{b_{n+1}}{b_n}$ must be constant.

Substituting $a_n=b_n+k$ into the recurrence relation for a_n:

$b_{n+1}+k=2(b_n+k)-3$

$b_{n+1}+k=2b_n+2k-3$

$b_{n+1}=2b_n+k-3$

For b_n to be a geometric sequence with common ratio r, we must have $b_{n+1}=rb_n$. Comparing this with the derived equation, we see that r must be 2, and the constant term $k-3$ must be zero.

So, $k-3=0$, which implies $k=3$. The common ratio r is 2.

Let's verify this. If $k=3$, then $b_n=a_n-3$.

$a_1=5$, so $b_1=5-3=2$.

$a_2=2a_1-3=2(5)-3=10-3=7$. So $b_2=7-3=4$.

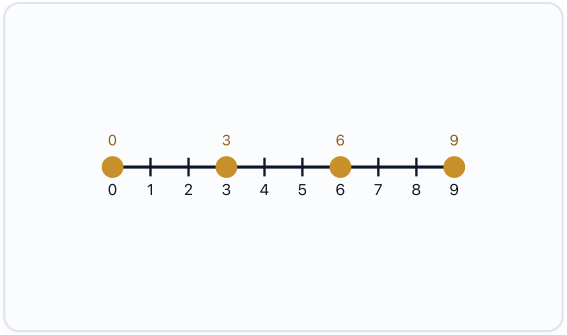
$a_3=2a_2-3=2(7)-3=14-3=11$. So $b_3=11-3=8$.

The sequence b_n starts with 2, 4, 8, ... This is a geometric sequence with first term 2 and common ratio $\frac{4}{2}=2$ or $\frac{8}{4}=2$.

Thus, $k=3$ and the common ratio $r=2$.

19 MCQ

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?



A -3

B 12

C 6

D 3 ✓

WORKED SOLUTION & VERIFICATION

The marked terms of the arithmetic sequence on the number line are $0, 3, 6, 9$. The common difference between consecutive terms is $3 - 0 = 3$.

20 MCQ

Find the common ratio of the geometric sequence $5,\ 10,\ 20,\ 40, \ldots$

A -2

B 3

C 5

D 2 ✓

WORKED SOLUTION & VERIFICATION

Divide consecutive terms: $10 \div 5 = 2$.

21 MCQ

Find the common ratio of the geometric sequence $5,\ -10,\ 20,\ -40, \ldots$

A 2

B -2 ✓

C -1

D -15

WORKED SOLUTION & VERIFICATION

Divide consecutive terms: $-10 \div 5 = -2$.

22 MCQ

Calculate the common ratio of the geometric sequence $4,\ -8,\ 16,\ -32, \ldots$

A -2 ✓

B 2

C -1

D -12

WORKED SOLUTION & VERIFICATION

Divide consecutive terms: $-8 \div 4 = -2$.

23 MCQ

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

- A 2
- B -2
- C -3
- D 3 ✓

WORKED SOLUTION & VERIFICATION

Let the first term of the geometric sequence be a_1 and the common ratio be r. \nThe sum of the first n terms of a geometric sequence is given by $S_n = a_1 \frac{1-r^n}{1-r}$ (for $r \neq 1$). \nWe are given that $a_1 = 2$. \nThe sum of the first three terms is $S_3 = 26$. So, $2\frac{1-r^3}{1-r} = 26$. \nThis simplifies to $\frac{1-r^3}{1-r} = 13$. \nUsing the difference of cubes formula, $1 - r^3 = (1 - r)(1 + r + r^2)$. \nThus, $1 + r + r^2 = 13$. \nRearranging this, we get a quadratic equation: $r^2 + r - 12 = 0$. \nFactoring the quadratic, we have $(r + 4)(r - 3) = 0$. \nThis gives possible values for r as $r = 3$ or $r = -4$. \nNow consider the sum of the first six terms, $S_6 = 728$. So, $2\frac{1-r^6}{1-r} = 728$. \nThis simplifies to $\frac{1-r^6}{1-r} = 364$. \nWe can rewrite $1 - r^6$ as $(1 - r^3)(1 + r^3)$. \nSo, $\frac{(1-r^3)(1+r^3)}{1-r} = 364$. \nWe know that $\frac{1-r^3}{1-r} = 13$. \nSubstituting this into the equation, we get $13(1 + r^3) = 364$. \nDividing by 13, we have $1 + r^3 = \frac{364}{13} = 28$. \nSo, $r^3 = 27$. \nThis implies $r = 3$. \nBoth conditions are satisfied when $r = 3$. \nIf we had chosen $r = -4$ from the first condition, $S_6 = 2\frac{1-(-4)^6}{1-(-4)} = 2\frac{1-4096}{5} = 2\frac{-4095}{5} = 2(-819) = -1638$, which is not 728. \nTherefore, the common ratio is 3.

24 MCQ

Calculate the common ratio of the geometric sequence $1,\ 4,\ 16,\ 64, \ldots$

- A -4
- B 4 ✓
- C 5
- D 3

WORKED SOLUTION & VERIFICATION

Divide consecutive terms: $4 \div 1 = 4$.

25 MCQ

Find the sum of the first 6 terms of the arithmetic sequence with $a_1 = 9$ and $d = 3$.

- A 105
- B 99 ✓
- C 144
- D 96

WORKED SOLUTION & VERIFICATION

$$S_6 = \frac{6}{2}(2(9) + (5)(3)) = 99.$$